

Paul Chapron

Computer Science, Ph.D.
Social Sciences Modeling and Simulation

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Current Position

since December 2017 **Researcher**, *LASTIG laboratory*, STRUDEL Team.

Scientific Interests

Main Themes Agent Based Modeling and Simulation, Complex Systems, Social and Spatial Simulation, Simulation Models Evaluation and Validation

Related Themes Sociology of Organizations, Quantitative Geography, Systems of Cities, Urban planning, Agricultural systems, Viability theory, Companion Modelling

Education and past Positions

- February 2017 Postdoc at the Institute Cmmplex Systems of Paris Ile de France (ISCPiF), CNRS UPS 3611. FUI Project Dynamicity. Calibration and robustness assessment of urban planning models. PI : Romain Reuillon.
- Oct-Dec2016 **Postdoc**, *French National Institute of Geography and Forestry Information (IGN)*, Exploration of morphological diversity of buildings according to Right to Build reglementation. Supervisor: Mickaël Brasebin and Julien Perret.
- 2015-2016 **Postdoc and Teaching Assistant**, *Institute of Geography and Durability, University of Lausanne*, Analysis and exploration of innovation geographical networks. Supervisor: Pr. Céline Rozenblat.
- 2013-2014 **Postdoc**, *Géographie-cités (UMR 8504 CNRS Paris I Paris VII)*, Modeling of system of cities evolution, within ERC grant project GeoDiverCity. Supervisor: Denise Pumain.
- 2007-2012 **PhD**, *Université Toulouse Capitole/IRIT UT1 UMR 5505, SMAC team*, Title: “Modélisation et analyse des organisations sociales: Propriétés Structurelles, régulation des comportements et évolutions”.Defended on May, 29th 2012.
- 2006-2007 **Research Master's degree**, *Université Henri Poincaré, Nancy*.
- 2003-2006 **Computer Science Engineering degree**, *ESIAL (now Telecom Nancy), Nancy*.

Teaching

2018-2012 **Teacher (vacataire), ENSG, Université Gustave Eiffel.**

Année Scolaire	Master 2 IGAST, ENSG, Université Gustave Eiffel	Mastère Spécialisé DE-SIGEO, ENSG, Université Gustave Eiffel	Total Heures
2018-2019	45h	9h	54h
2019-2020	42h	9h	51h
2020-2021	51h	12h	63h
2021-2022	50h	37h	87h

Softwares : R , QGIS

Topics: Statistics introduction: bivariate and descriptive statistics, Spatial analysis student project management.

2015-2016 **Teaching Assistant, University of Lausanne, IGD.**

Involved in the **MOOC** "*Analysis, Visualization and Simulation of Geographical Networks*" of Pr. Céline Rozenblat. Video courses, courses materials and tutorials production.

Topics : Clustering techniques, Random network and ERGM models, Explanatory analysis of networks, Simple network dynamics simulation.

Softwares : TULIP (visualization), R (analysis and modeling), Netlogo (simulation)

2007-2012 **Teaching Assistant, University of Toulouse Capitole.**

Office software training	Bachelor students
Information Systems	Master students
Computer Architecture	Master students
Social Systems Modeling and Simulation	Master students

2006-2007 **Contractor Teaching Assistant, Université de Nancy II.**

Office software training	Bachelor students
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PhD supervision

2021-2024 **Jonathan Siliézar-Montoya, UMRAE, Acoustics, Defended the 06/12/2024.**

Multi-source modelling of urban sound environments

Internships supervision

2021 **Master Student, Ludovic Moisan, Master 2.**

Exploration des données issues de l'application NoiseCapture, par le prisme des sources sonores

2021 **Master Student, Jonathan Siliézar-Montoya, Master 2.**

Multi-sources modelling of urban acoustic landscapes(Projet LASSO, I-site FUTURE 2020)

2020 **Master Student, Luke Riley, Master 2.**

Joint characterization and visualization of urban climate and morphology

2018 **Master Student, Vincent Delbar, Master 2.**

Urban sprawl simulation scenarios in Occitanie 2040

Peer-reviewed Articles

International Journal Articles

- [1] P. Chapron, M. Leclaire, A. Scriban, D. Masse, R. Reuillon, A. Fallot, and E. Delay. Fallow land simulations to fight against soil degradation: Engaging stakeholders with model exploration. *Journal of Artificial Societies and Social Simulation*, 29(1):3, 2026. ISSN 1460-7425. doi: 10.18564/jasss.5816.
- [2] J. Siliézar, P. Aumond, A. Can, P. Chapron, and M. Péroche. Case study on the audibility of siren-driven alert systems. *Noise Mapping*, 10(1), 2023. doi:10.1515/noise-2022-0165.
- [3] K. Widyastuti, R. Reuillon, P. Chapron, W. Abdussalam, D. Nasir, M. E. Harrison, H. Morrogh-Bernard, M. A. Imron, and U. Berger. Assessing the impact of forest structure disturbances on the arboreal movement and energetics of orangutans—an agent-based modeling approach. *Frontiers in Ecology and Evolution*, Volume 10 - 2022, 2022. ISSN 2296-701X. doi:10.3389/fevo.2022.983337.
- [4] C. Cottineau, P. Chapron, and R. Reuillon. Growing models from the bottom up. an evaluation-based incremental modelling method (ebimm) applied to the simulation of systems of cities. *Journal of Artificial Societies and Social Simulation*, 18(4):9, 2015. ISSN 1460-7425. doi: 10.18564/jasss.2828.
- [5] C. Cottineau, R. Reuillon, P. Chapron, S. Rey-Coyrehourcq, and D. Pumain. A modular modelling framework for hypotheses testing in the simulation of urbanisation. *Systems*, 3(4):348, 2015. ISSN 2079-8954. doi:10.3390/systems3040348.
- [6] C. Sibertin-Blanc, P. Roggero, F. Adreit, B. Baldet, P. Chapron, J. El-Gemayel, M. Mailliard, and S. Sandri. Soclab: A framework for the modeling, simulation and analysis of power in social organizations. *Journal of Artificial Societies and Social Simulation*, 16(4):8, 2013. ISSN 1460-7425.
- [7] P. Chapron. Analyse structurelle des champs de capacité d'action dans les organisations sociales. outils analytiques pour la caractérisation de la coopération. *Studia Informatica Universalis*, 10(1):98–128, 2012.
- [8] P. Chapron. Analyse de reseau de pouvoir au sein d'une organisation sociale. *Nouvelles Perspectives en Sciences Sociales*, 6(2):233–256, 2011.

French Journal Articles

- [9] Delay, Etienne, Chapron, Paul, Leclaire, Mathieu, Reuillon, Romain, and Faye, Ousmane. Entre terre et solidarité: la jachère collective comme lien social à dioline, sénégal. *Cah. Agric.*, 34:28, 2025. doi:10.1051/cagri/2025029.
- [10] J. El Gemayel, P. Chapron, F. Adreit, and C. Sibertin-Blanc. Quand et comment les acteurs sociaux peuvent-ils coopérer ? Un algorithme de simulation pour la négociation de leurs comportement. *Revue d'Intelligence Artificielle*, 25, 2011.
- [11] C. Sibertin-Blanc, F. Adreit, P. Chapron, J. El Gemayel, M. Mailliard, P. Roggero, and C. Vautier. Compte-rendu d'une recherche interdisciplinaire entre sociologues et informaticiens : de la sociologie de l'action organisée au logiciel SocLab. *Technique et Science Informatiques*, 29(8-9):1081–1115, 2010.

International Conference Proceedings

- [12] Paul Chapron and Christophe Sibertin-Blanc. Analysis of the power network among the actors of a social organization (regular paper). In D. Kazakov and G. Tsoulas, editors, *AISB Symposium on Social Networks and Multiagent Systems, York, 04/04/2011-07/04/2011*, pages 2–7, <http://www.aisb.org.uk>, avril 2011. Society for the Study of Artificial Intelligence and the Simulation of Behaviour.

- [13] Joseph El Gemayel, Christophe Sibertin-Blanc, and Paul Chapron. Impact of tenacity upon the Behaviors of Social Actors (regular paper). In Alexis Drogoul, Benoit Gaudou, and Nicolas Marilleau, editors, *Applied Agent based simulator Engineering for Complex System study, Nagoya, Japan, 13/12/2009*, 2009.

Book Contributions

- [14] C. Cottineau, P. Chapron, M. L. Texier, and S. Rey-Coyrehourcq. Incremental Territorial Modelling. In *Geographical Modeling: Cities and territories*, 1. Wiley, 2020. doi:10.1002/9781119687290.ch4.
- [15] C. Cottineau, P. Chapron, M. L. Texier, and S. Rey-Coyrehourcq. Modélisation territoriale incrémentale. In D. Pumain, editor, *La modélisation en géographie. Villes et territoires, Méthodologies de modélisation en sciences sociales*. ISTE, 2020.
- [16] C. Paul, C. Guillaume, C. Clémentine, C. Robin, L. Mathieu, P. Denise, R.-C. Sébastien, R. Romain, S. Clara, and S. Elfie. *Urban Dynamics and Simulation Models*. Springer International Publishing, 2017. doi:10.1007/978-3-319-46497-8.
- [17] F. Adreit, P. Chapron, J. El GEmayel, P. Roggero, S. Sandri, and C. Sibertin-Blanc. Une formalisation de la sociologie de l'action organisée : métamodèle, simulation et étude analytique. In D. Phan, editor, *Ontologies et modélisation par SMA en SHS (informatique et systèmes d'information, rta)*. Hermes Science Publications, 2014. ISBN 9782746232075.
- [18] J. El Gemayel, C. Sibertin-Blanc, and P. Chapron. Impact of Tenacity upon the Behaviors of Social Actors. In Q. Bai and N. Fukuta, editors, *Advances in Practical Multi-Agent Systems*, volume 325, pages 287–306. Springer, <http://www.springerlink.com>, 2011.

Other Publications

Workshop and Communications

- [19] C. Cottineau and P. Chapron. Evaluation & calibration for the comparison of abms of cities' trajectories, 2014. Paper Session: 2539 Geosimulation Models 3: Applications - Macro,AAG Meeting 2014, Tampa (Florida) <http://meridian.aag.org/callforpapers/program/SessionDetail.cfm?SessionID=19892>.
- [20] P. Chapron. Analysing the dynamics of a social organization by means of its transfer function, 2013. International Conference "Interactions in Complex Systems" and 12th International Workshop "Spatial Econometrics and Statistics", June 17-19, Université d'Orléans <http://www.univ-orleans.fr/mapmo/peps/?q=content/international-conference-interactions-complex-systems-and-12th-international-workshop> http://www.univ-orleans.fr/mapmo/peps/sites/default/files/conferences/programme_ISC_Print.pdf.
- [21] P. Chapron. Diversité des formes d'îlots urbains à partir d'un Plan Local d'Urbannisme, 2013. Journée de la recherche IGN, 23 mars 2017. <http://recherche.ign.fr/jr/jr17.php>.
- [22] C. Cottineau and P. Chapron. Back in the USSR: a series of models for simulating urban evolution, 2013. Urban Systems Modeling: advances from GeoDiverCity and MECHANICITY ERC programmes. Satellite meeting of the European Conference on Complex Systems 2013, Barcelona (Spain).
- [23] C. Cottineau and P. Chapron. Step-wise modeling. An Incremental method for modeling cities trajectories in the post-Soviet urban system., 2013. 18th European Colloquium in Theoretical and Quantitative Geography (ECTQG) .
- [24] P. Chapron and J. El Gemayel. Social Actors Rationality within Organizations, 2009. Simulation of Complex Social Systems, (conférence sans actes) Sevilla, Spain, 09/11/2009.

- [25] F. Adreit, P. Chapron, , C. Sibertin-Blanc, P. Roggero, and S. Sandri. Une ontologie des systèmes d'actions concrets, 2008. Rencontres interdisciplinaires sur les systèmes complexes naturels et artificiels - Journées de Rochebrune, Rocchebrune, 19/01/09-23/01/09.
- [26] F. Adreit, P. Chapron, P. Roggero, and C. Sibertin-Blanc. Régulation des organisations et rationalité des acteurs, 2008. XVIII Congrès de l'AISLF, Istanbul (conférence sans actes), 07/07/08-11/07/08.